EXPERIMENTAL DESIGN (MDA3063)

2^k FACTORIAL DESIGN

GROUP 4

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GENERAL INTRODUCTION OF FACTORIAL DESIGN



Definition: Tools to study effects of multiple factors simultaneously



Allows researchers to examine how different factors interact with each other



Efficient approach for multi-factor experiments

WHAT IS FACTORIAL DESIGN?

2^K FACTORIAL DESIGN

- K = NUMBER OF FACTORS IN THE EXPERIMENT
- 2 = TWO LEVELS PER FACTOR (HIGH AND LOW)
- NOTATION:

  HIGH LEVEL = +1

  LOW LEVEL = -1
- PURPOSE: EXPLORE LARGE NUMBER OF FACTORS WITH MINIMAL LEVELS

2^3 FACTORIAL DESIGN

- INVOLVES THREE FACTORS (K = 3)
- EACH FACTOR TESTED AT TWO LEVELS (HIGH AND LOW)
- EXAMINES ALL POSSIBLE COMBINATIONS OF THE THREE FACTORS WITH TWO REPLICATION (16 RUNS)

OBJECTIVE

The main objective for this study is to investigate the effects of three factors with 2 replication (16 runs) : oven temperature, baking time and flour type on cake volume (cm^3).

FACTORS AND LEVELS :

Factors	Low Level (-1)	High Level (+1)
Oven Temperature (A)	160°C	180°C
Baking Time (B)	30min	40 min
Flour Type (C)	All-purpose flour	Cake flour

EXPERIMENTAL DESIGN MATRIX

Run	Oven Temperature (A)	Baking Time (B)	Flour Type (C)	Treatment Comb.	Replicate Volumes (cm³)
1	−	−	−	(1)	850, 845
2	+	−	−	a	920, 915
3	−	+	−	b	880, 875
4	+	+	−	ab	950, 945
5	−	−	+	c	870, 865
6	+	−	+	ac	930, 925
7	−	+	+	bc	900, 895
8	+	+	+	abc	970, 965

ANOVA MODEL : $Y = \mu + A + B + C + AB + AC + BC + ABC + \varepsilon$

MAIN EFFECTS

Main Effect of A:

$\frac{1}{4(2)}$ (917.5 + 947.5 + 927.5 + 967.5 - 847.5 – 877.5 – 867.5 – 897.5)

= 33.75

Main Effect of B:

$\frac{1}{4(2)}$ (947.5 + 877.5 + 897.5 + 967.5 – 847.5 – 917.5 – 867.5 – 927.5)

= 16.25

Main Effect of C:

$\frac{1}{4(2)}$ (867.5 + 927.5 + 897.5 + 967.5 – 847.5 – 917.5 - 877.5 – 947.5)

= 8.75

INTERACTION EFFECTS

Interaction Effect of AB:

$\frac{1}{4(2)}$ (847.5 + 947.5 + 867.5 + 967.5 – 917.5 – 877.5 – 927.5 – 897.5)

= 1.25

Interaction Effect of AC:

$\frac{1}{4(2)}$ (847.5 + 877.5 + 927.5 + 967.5 – 917.5 – 947.5 – 867.5 – 897.5)

= – 1.25

Interaction Effect of BC:

$\frac{1}{4(2)}$ (847.5 + 917.5 + 897.5 + 967.5 – 877.5 – 947.5 – 867.5 - 927.5)

=1.25

Interaction Effect of ABC:

$\frac{1}{4(2)}$ (967.5 + 867.5 + 877.5 + 917.5 – 847.5 – 947.5 – 927.5 – 897.5)

=1.25

RESULT

- As we compare , the regression coefficient is one-half the effect estimate.
- This confirms our manual calculations on the main and interaction effect .

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Effect A	67.5000000	1.76776695	38.18	<.0001
Effect B	32.5000000	1.76776695	18.38	<.0001
Effect C	17.5000000	1.76776695	9.90	<.0001
Effect AB	2.5000000	1.76776695	1.41	0.1950
Effect AC	-2.5000000	1.76776695	-1.41	0.1950
Effect BC	2.5000000	1.76776695	1.41	0.1950
Effect ABC	2.5000000	1.76776695	1.41	0.1950

PARAMETER	Manual Calculation
Effect A	33.75
Effect B	16.25
Effect C	8.75
Effect AB	1.25
Effect AC	−1.25
Effect BC	1.25
Effect ABC	1.25

MAIN EFFECTS HYPOTHESIS

FACTOR A
H_0A : FACTOR A HAS NO EFFECT ON THE RESPONSE VARIABLE.
H_1A : FACTOR A HAS A SIGNIFICANT EFFECT ON THE RESPONSE VARIABLE.

FACTOR B
H_0B: FACTOR B HAS NO EFFECT ON THE RESPONSE VARIABLE.
H_1B: FACTOR B HAS A SIGNIFICANT EFFECT ON THE RESPONSE VARIABLE.

FACTOR C
H_0C: FACTOR C HAS NO EFFECT ON THE RESPONSE VARIABLE.
H_1C: FACTOR C HAS A SIGNIFICANT EFFECT ON THE RESPONSE VARIABLE.

TWO-WAY INTERACTION HYPOTHESIS

INTERACTION A × B
H_0AB: THERE IS NO INTERACTION EFFECT BETWEEN FACTORS A AND B.
H_1AB: THERE IS A SIGNIFICANT INTERACTION EFFECT BETWEEN FACTORS A AND B.

INTERACTION B × C
H_0BC: THERE IS NO INTERACTION EFFECT BETWEEN FACTORS B AND C.
H_1BC: THERE IS A SIGNIFICANT INTERACTION EFFECT BETWEEN FACTORS B AND C.

INTERACTION A × C
H_0AC: THERE IS NO INTERACTION EFFECT BETWEEN FACTORS A AND C.
H_1AC: THERE IS A SIGNIFICANT INTERACTION EFFECT BETWEEN FACTORS A AND C.

THREE-WAY INTERACTION HYPOTHESES

INTERACTION A × B × C
H_0ABC: THERE IS NO THREE-WAY INTERACTION AMONG FACTORS A, B, AND C.
H_1ABC: THERE IS A SIGNIFICANT THREE-WAY INTERACTION AMONG FACTORS A, B, AND C.

ANOVA RESULT

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Analysis of Variance Table

Response: volume
           Df Sum Sq Mean Sq F value    Pr(>F)
A           1  18225 18225.0    1458 2.430e-10 ***
B           1   4225  4225.0     338 7.888e-08 ***
C           1   1225  1225.0      98 9.153e-06 ***
A:B         1     25    25.0       2     0.195
A:C         1     25    25.0       2     0.195
B:C         1     25    25.0       2     0.195
A:B:C       1     25    25.0       2     0.195
Residuals   8    100    12.5
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
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- All the main effect have p-values less than the significance level (α = 0.05). So we reject HO for the main effects
- Indicating that Oven Temperature, Baking Time, and Flour Type significantly affect the volume of the cake.
- All two-way and three-way interaction effects have p-values greater than 0.05. Hence,we failed to reject for the interaction effects
- There is no significant interaction between Oven Temperature, Baking Time, and Flour Type on the cake volume.

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thank you